Exercise II

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1 Gambles and Lower Previsions

1.1 Probability Intervals

Consider a random variable X with a finite outcome space $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$. You are told that the probabilities of each elementary event are not known precisely, but lie within the following interval constraints.

Outcome	**Lower probability** (l_i)	**Upper probability** (u_i)
ω_1	0.1	0.3
ω_2	0.2	0.4
ω_3	0.1	0.5
ω_4	0.0	0.3

The only global constraint is the probabilities must sum to one: $\sum_{i=1}^{4} P(\omega_i) = 1$.

(a) List the set of probability distributions that are all compatible with the information we have.

(b) How many free variables does this set have?

(c) What is the geometric shape of this set in $\mathbb{R}^4$?

(d) Compute the lower and upper probabilities of the event: $\{\omega_1, \omega_3\}$.

(e) Verify the coherence property: $\underline{P}(A) \leq \overline{P}(A)$ for all events A.

1.2 Avoiding Sure Loss and Coherence

A football analyst considers the result of a match, $\Omega = \{W, D, L\}$ (win, draw, loss). She states her supremum buying prices for four indicator gambles:

gamble f	$\mathbf{1}_W$	$\mathbf{1}_D$	$\mathbf{1}_L$	$\mathbf{1}_{\{W,D\}}$
$\underline{P}(f)$	0.3	0.2	0.4	0.4

Let $\mathcal{K}$ be the set of these four gambles. Recall from the lecture that $\underline{P}$ *avoids sure loss* if

$$\sup_{\omega\in\Omega} \sum_{i=1}^{n} \big[f_i(\omega) - \underline{P}(f_i)\big] \geq 0 \tag{1}$$

for all $n \geq 0$ and all $f_1, \ldots, f_n \in \mathcal{K}$. It is *coherent* if

$$\sup_{\omega\in\Omega} \Big\{ \sum_{i=1}^{n} \big[f_i(\omega) - \underline{P}(f_i)\big] - m\big[f_0(\omega) - \underline{P}(f_0)\big] \Big\} \geq 0 \tag{2}$$

for all $n, m \geq 0$ and all $f_0, f_1, \ldots, f_n \in \mathcal{K}$. The gambles f_i need not be distinct.

(a) Using the definitions of coherence and avoiding sure loss, show that every coherent lower prevision avoids sure loss.

(b) Show that the converse is not true by demonstrating that the analyst's $\underline{P}$ is not coherent. *Hint:* Take $f_0 = \mathbf{1}_{\{W,D\}}$. Choose m and gambles $f_1, \ldots, f_n$ from the other three so that the expression inside the supremum is negative for every ω. Rearrange your inequality to show that the analyst should accept to buy $\mathbf{1}_{\{W,D\}}$ for more than 0.4.

(c) Now suppose the numbers instead of $\underline{P}$ represent the fair prices at which the analyst both buys and sells the gambles. Construct a set of gambles to show that now incoherence causes sure loss to the analyst.

(d) Draw the credal set $\mathcal{M}(\underline{P}) = \{p : \mathbb{E}_p[f] \geq \underline{P}(f) \text{ for all } f \in \mathcal{K}\}$ in the probability simplex and find its vertices. Compute the lower envelope $\underline{E}(f) = \min_{p\in\mathcal{M}(\underline{P})} \mathbb{E}_p[f]$ for the four gambles in $\mathcal{K}$ and for the gamble $g = (g(W), g(D), g(L)) = (2, -1, 0)$. Which prices change, and how does this relate to (b)?

2 Possibility Theory

- Coding exercise: `https://colab.research.google.com/drive/1qEs8zWAUK6dwAxWldJ3Xuhpkfa9NX-6f?usp=sharing`